UMD DATA605 - Big Data Systems

# 12.1: Graph Data Management


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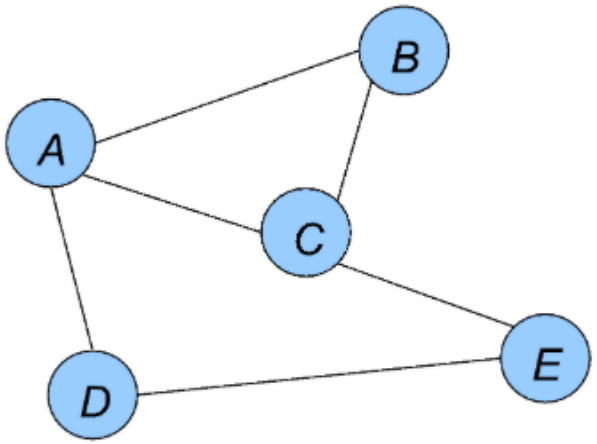
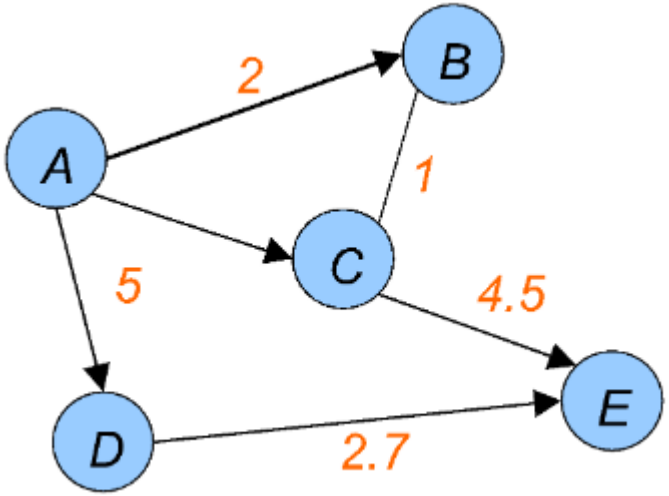
# Graphs: Background

- A **graph** represents entities and their connections
  - *Entities* are vertices (nodes)
  - *Connections* are edges (links, arcs, relationships)
- **Applications of graphs in many fields**
  - Social networks
  - Biological networks
  - Information networks
  - Infrastructure networks
  - ...



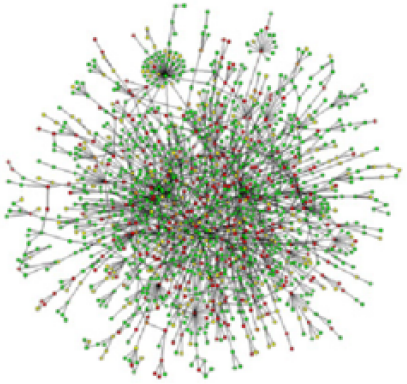

*An undirected, unweighted graph*



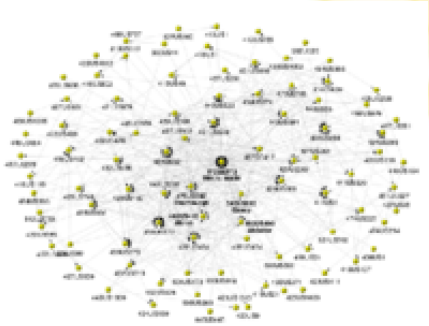

*A directed, edge-weighted graph*

# Graph Data Structures: Motivation

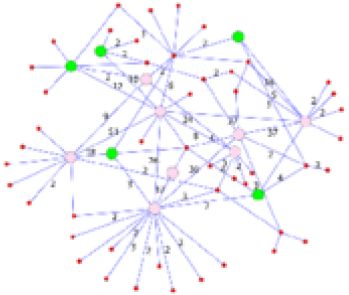
- **Graph data**:
  - Increasing volumes of data
  - Increasing interest in querying and reasoning
- **Sectors**
  - Healthcare
  - Finance
  - Logistics
- **Example applications**
  - Fraud detection
  - Recommendation systems
  - Network analysis



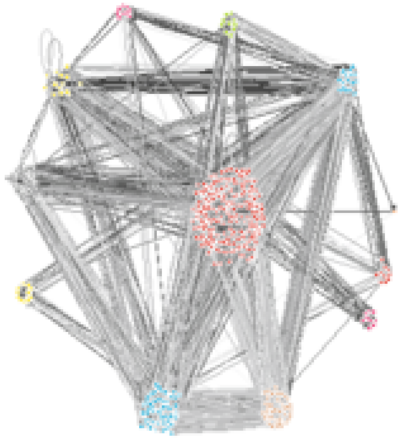
*Protein-protein interaction network*



*Stock trading network*



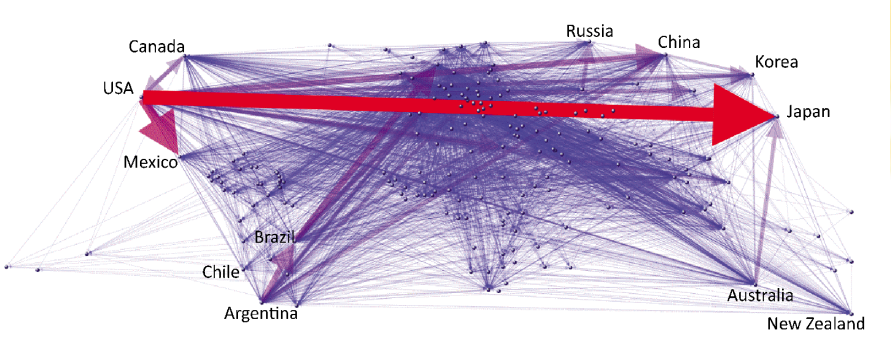
*Supreme court citation network*



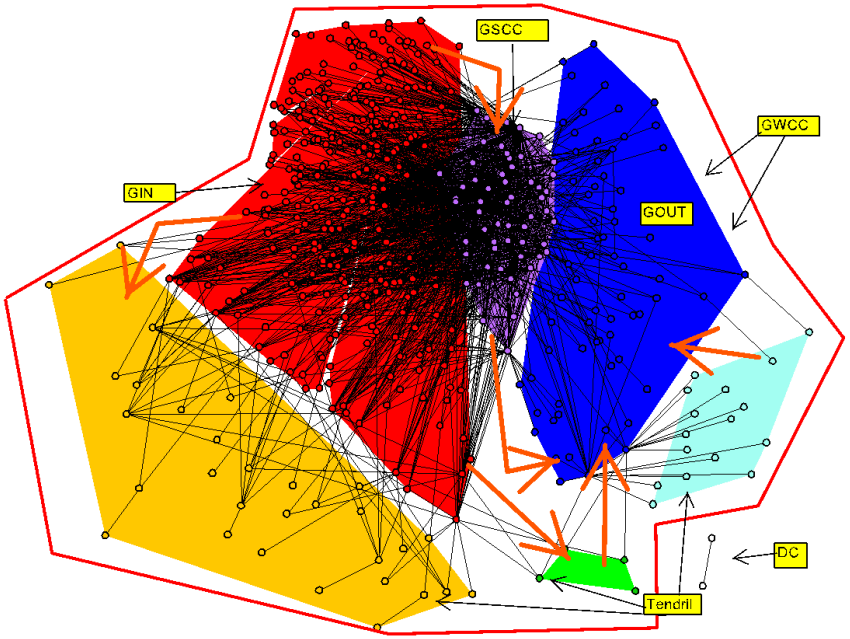
*Social networks*

# Graph Data Structures: Motivation

- **Traditional tools** (e.g., relational DBs, NoSQL DBs) struggle with:
  - Storing and querying graph data
  - Processing graph-structured queries
- **Dedicated solutions** to:
  - Storing: Neo4j
  - Processing: Google Pregel, Apache Giraph, Spark GraphX




*Global virtual trade network*



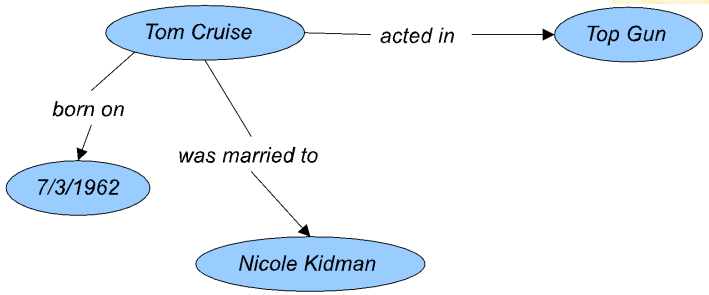

*Federal funds networks*

# Knowledge Graphs

- **Representation of knowledge in the form of graphs**
  - Capture entities, relationships, properties
  - Provide structured view of real-world information
  - E.g., Google Knowledge Graph, DBpedia, Wikidata
  - E.g., RDF or Property Graph models
- **Applications**
  - Enable machine understanding of complex domains
  - Support semantic search, recommendation, analytics
  - Used in industries for data integration, knowledge discovery, AI applications
- **Ontologies**
  - Provide formal and structure representation of knowledge
    - E.g., type of things and how they relate to each other
    - Like "meaning and rules" to interpret data into a knowledge graph
  - Promote interoperability across knowledge bases

# Graph Data Models: RDF

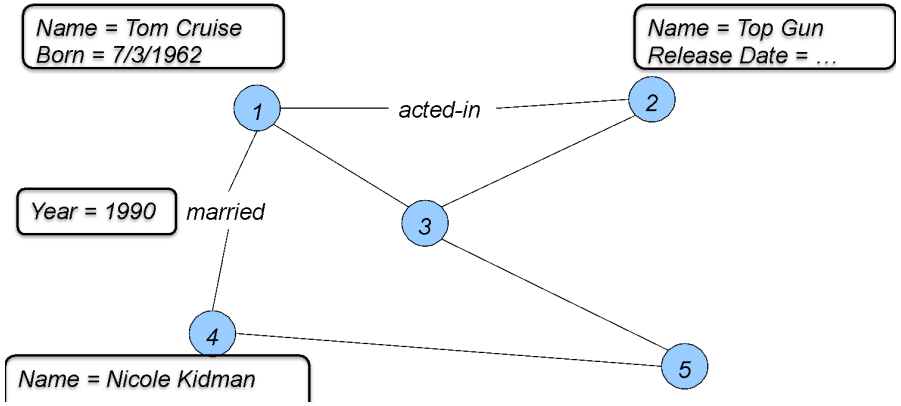
- **Resource Description Framework**
  - RDF uses triples: subject-predicate-object
  - Connects "subject" and "object" through "predicate"
  - E.g., "TomCruise-acted-TopGun"
- **Used to represent knowledge bases**
  - Queried through SPARQL




- **Pros**
  - Standardization
    - W3C standard to model data
    - Subject and object can be URIs in semantic web
  - Interoperability
    - Merge RDF data stores
  - Extensibility
    - Add new nodes and relationships
    - Support ontologies

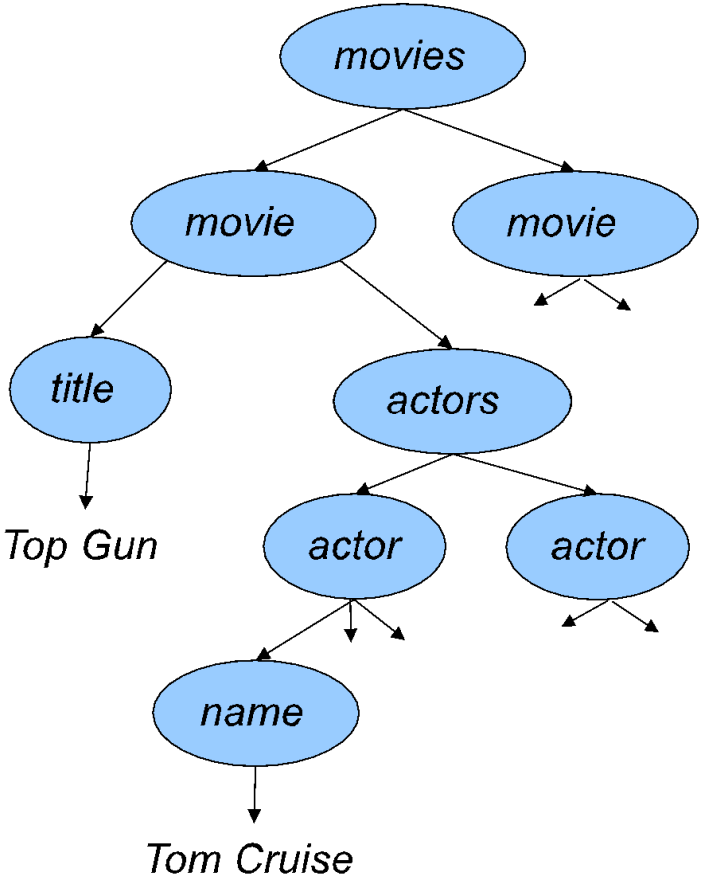
# Graph Data Models: Property Graph

- Directed graph with nodes and edges having key-values *properties*
  - Similar expressive power to RDFs
- **No universal standard**
  - Less "schema"
  - Harder to interoperate
- **Examples of query languages**
  - Cypher for Neo4j
  - Gremlin for Apache TinkerPop



# Graph Data Models: XML

- Common data model for flexible data representation
  - Directed labeled tree
  - Popular for non-tabular data exchange

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  <movie>
  <title>Top Gun</title>
  <actors>
    <actor>
     <name>Tom Cruise</name>
     <born>7/3/1962</born>
    </actor>
   <actor>
     ...
   </actor>
  </actors>
  </movie>
```

# Comparison

| Feature | RDF | Property Graph | XML |
|---|---|---|---|
| Core data model | Triples: **(subject, predicate, object)** | **Nodes** and **edges** with properties | **Hierarchical tree** of elements |
| How facts are stored | Fact is a separate triple | Facts are properties on nodes or edges | Nested tags with attributes |
| Attributes | Modeled as triples | Stored as key–value pairs | XML attributes or child elements |
| Semantics | Formal semantics (RDF, RDFS, OWL) | No built-in semantics | No built-in semantics |
| Ontology support | Native and standardized | Optional, application-defined | None (only schema) |
| Reasoning & inference | Built-in (e.g., OWL/RDFS) | Usually not supported | Not supported |
| Data integration | Excellent for heterogeneous data | Requires manual mapping | Difficult across different schemas |
| Query language | `SPARQL` | Cypher, Gremlin, GQL | XPath, XQuery |
| Query style | Graph pattern matching | Graph traversal patterns | Tree navigation |
| Schema | Optional but strongly supported | Labels and constraints | XSD, DTD |
| Standards | W3C standards | Vendor-specific | W3C standards |
| Interoperability | Very high | Limited | High for doc exchange |
| Performance for traversals | Moderate | Very fast | Poor for graph traversal |
| Typical use cases | Linked data, knowledge bases | Fraud, recommendations, network analysis | Doc storage, configuration, data exchange |
| Example systems | Wikidata, DBpedia, Apache Jena | Neo4j, Amazon Neptune | SOAP, RSS, Office documents |

# Storing Graph Data

- **File systems**
  - Simple
  - No transactions, ACID compliance
  - Minimal functionality (build analysis/querying on top)
- **Relational databases**
  - Mature technology
  - SQL, transactions, ACID compliance, toolchains
  - Minimal functionality for graph data
- **NoSQL key-value stores**
  - Handle large datasets efficiently in a distributed fashion
  - Minimal native functionality for graph data
- **Graph databases**
  - Efficiently support complex queries/tasks (e.g., graph traversals)
  - Less mature than RDBMSs
  - Often lack declarative language like SQL
    - Write custom programs

# Graph Databases

- Many specialized graph DB systems
  - E.g., Neo4j, Titan, OrientDB, AllegroGraph
- Key distinctions from relational / NoSQL databases
  - Manage and query graph-structured data
  - Store graph structure with pointers
    - Avoid joins, simplify graph traversals
    - Write queries and graph algorithms (reachability, shortest paths)
  - Support graph query languages: SPARQL, Cypher, Gremlin
  - Rudimentary declarative interfaces
  - Applications require programmatic interfaces
  - Provide programmatic API for arbitrary graph algorithms